ActInf Textbook Group ~ Cohort 7 ~ Session 23 (Chapter 9) ~ 4/10/2025

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Hey, welcome fellows. Um, we're in the
second discussion on chapter 9 for
cohort 7. So, we could go to any part of
chapter
9 or jump in wherever you prefer.
What about the provocative first
sentence? Ultimately, the models
described in this book are only useful
if they can answer scientific questions.
Uh yeah, I liked uh this is uh old
chapter because of its
uh uh well because of the way that it um
inverts everything for the sake of uh uh
applying this to science. I think that
that's it's interesting the cognizance
of it all to realize like uh maybe
what's been done so far is helpful as a
model but that doesn't really um uh
that's not the same as let's say a
scientific
um I guess result and so but to turn it
around and say it is quite impressive if
you can uh do it backwards so that if
you can if I understood this correctly
If you can suppose that there's this
model in a subject or maybe a group of
subjects and then you can um given that
supposition then maybe figure out what
the parameters should be
um or what the what the results should
be and then work back to say okay well
if the if these are the results then
what are the initial beliefs you know
between different um participants in a
population and then if you can do that

then it's interesting oh that you have a
a you know potentially a scientific
explanation of how these different uh
individuals in a population differ and
why and to be able to show the
helpfulness of the model and as
explanatory and then you can correlate
that with other information. So that's
how what I took from
this. Yeah, great points. It has a lot
to do with what makes a scientific
account like any let alone a satisfying
or useful one. And what is the
relationship between empirical
observables like the brute fact of the
data points and inferred latent states
like causal attributions.
Mhm. or unobserved features which are
derived from empirical data but derived
from empirical data does a lot of work
because you could
derive in in so many different
directions from empirical data. You
could say that this art piece was
derived from the weather data and
there's so many degrees of freedom with
what weather data you took and and what
art piece you resulted in that somebody
else might not be able to look at it and
know that you use weather data at all.
Right
now, I'm interpreting this statement to
mean
uh I'm interpreting answer scientific
questions to mean make testable
predictions. That may not
be everyone's interpretation, but I
always thought models are only useful if

they can make testable predictions.
Yeah, that's a great point that's
related to um like
falsificationism and the idea that we
create
secondary hypotheses that could then be
either assessed to be true or false
under the logic like if what you learned
from the
answer didn't give any derivative
product that could be deemed to be more
or less
valid. It was kind of a dead end
answer because it doesn't it doesn't
give any uh
consequences. And then testable
predictions is like the empiricist way
of framing it and that can even be
framed and staying within the
actontology like answering scientific
questions means reducing uncertainty
about the scientific question.
Oh, I like that. Now, alternatively,
um, pointing at your the first part of
what you just said, it's still useful if
we're moved forward in our
investigations, even if it doesn't have
explicitly testable
predictions. If we're still developing
the field of inquiry, that's still
valuable.
And and uh when you say testable
predictions, I think that that comes at
it from both ends that are necessary in
science because you want to have things
related to data that's observed, you
know. So if it's if it if there's no
data involved and there's no

observations involved, then that's a
funky science. But then also you want to
have a theory. Uh so you want to have a
a model that is um meaningfully breaking
down or decomposing or structuring um
these variables or questions or ideas
and so that you can look at
relationships and so if you have the
testable prediction it means that you
have it from both ends and then you can
apply it to a wider scope you know you
can look at new examples etc but I think
in the case of this chapter it's about
predictions of predictions which is very
um sophisticated because you're
basically predicting
um well that you can figure out what's
in another organism's mind, you know,
what they're predicting, let's say, and
then and if you're if your predictions
correct, you'll be able to do that for a
whole variety of individuals to say,
well, they're predicting these different
things. And you'll be able to confirm
that uh and then so the confirmation can
also happen on both levels that you can
do that for more and more individuals
but given an individual you can also get
better and better validation for like
well what exactly is that uh belief or
prediction that they're making. Um so
it's sophisticated again pointing to the
process nature the dynamic
nature the approach
that and and talking about it this way
with chapter 9 being so grounded with
empirical data it's kind of like the the
mirror symmetry of the

book. Chapter 1 and 10 are like these overview chapters. Chapter two is the low road. So when we when we when we got on that on-ramp, it was about building the model and the joint distribution and basian statistics. And now we're kind of exiting on an empirical low road because it's again reconnecting us back to the observations. Like this is a setting that's very familiar to a behavioral scientist. And even when a paper doesn't uh like a behavioral science paper doesn't explicitly use these techniques doesn't use a palm dp doesn't calculate variational free energy still basically the setting of this metabasian um setup is there like here it's the temperature and the humidity in the laboratory and they're the uh experimentter is playing sequences of videos or they're presenting symbols on cards which are observations for the subject and then they're receiving the action output of the subject. And then this just does so much interesting work with the subjective because it is both subjective to them and it's the subject of the behavioral experiment. And it's like this is what a laboratory is like. And then this solid box is like the kind of the veil of unknowing for the behavioral

scientist in terms of they can set up
what's in the room. They can pass the
stimuli to the subject and they can
observe what comes out with biometrics
or or with whatever kind of
measure and then what's inside is
proposed.
So this
structure it is is a really
parsimmonious way to just look at the
behavioral scientific
setting. And it turns out it has a lot
of structural congruence with the
metaccognitive
phenomenology like an observer
observing with within an
[Music]
organism. But that goes into kind of a
different direction than this
like more behavioral laboratory
very practical and and this and this
does get really
practical and the by number or by
citation count most of Fristan's papers
may be this kind of like this many
subjects were put into the fMRI for this
many minutes exposed to these stimuli
here was the statistical power here were
the the um here was how the experiment
happens.
Um, so this gets to the most
practical statistical power calculation
like parts of doing behavioral
science which kind
of in in the um
skarian area like BF Skinner that
without going too much in this that's
kind of the whole debate. Does that deny

mind? Does behaviorism exclude mind?
Um, and the the the cover of the
book told us that we were going to be in
for mind, brain, and
behavior. And we saw brain in chapter 5
with the different neural
systems. Chapter 9 really goes after
behavior.
because it's not making psychological
assertions
um or or
um philosophical
assertions. So mind is obviously a a
really complex and rich
area with the first person and the
second and the third person all these
and all these different traditions and
and our own uh firsthand experiences of
mind.
Uh but nine zeros in on
behavior which in a way gives the least
contentious of the three. That statement
might be contentious to some but the
neural accounts somebody might think
well with um molecular reductionism and
genetics and systems biology and deep
learning we're doing better than chapter
5 on brain accounts.
But in terms of the structure of the
uh
environment, this is really like a gem
tucked in at the end of the
book. I like that perspective. Uh there
are many traditions that say reality is
what happens. That's behavior.
Well, and I think for this chapter, I
don't know if I'd entirely agree, but
maybe I understand u but with Daniel,

but uh uh because a lot of these uh
examples at the end have to do with
psychological um analyses like let's say
hallucinations, right? Or delusions,
right? So that's partly behavioral, but
that's
partly their internal models, right?
like and and I think one of the
challenges with these experiments just
thinking about u what it would be like
to do one is that with active inference
in its full glory it has u two outcomes
you could be updating your environment
or you could be updating your belief so
then if you're doing an experiment you
want to capture that I think that'd be
like the whole glory of it you know you
have to be on top of two different uh
channels you know the external like the
the behavior you to be able record, oh,
you know, behavior updated the
environment, let's say, but you also
want to be able to capture, well, oh,
here they updated their
belief. So, that already is kind of like
a double thing. I don't know if that's
I'm
great great points and and and
challenging ones. So yeah from the first
half of the book we have this two ways
to to dissipate free energy with update
behavior or update belief. Now from the
behavioral uh researchers
perspective everything coming out of
that box is observed behavior whether
it's an EEG trace
um or whether it's a self-report or
whether it's a motor behavior. So from

the outside everything that we're
observing is behavioral and basically
everything that is internal is being
modeled as a belief.
That doesn't mean that it's like a um
propositionally held sentiment. Mhm. But
it's being modeled as a basian belief
which undergoes updating. So you're
totally right that if we want the full
active inference
account, we do want to be able to
describe both the updating of behavior,
but this is the updating of observables.
So if we only wanted to describe
behavioral updating, we could just do
descriptive statistics on sequences of
actions. And there might be a situation
where that's totally fine. But as soon
as the behavioral researcher starts to
suppose some latent
factor that wasn't behaviorally
observed but is um abduced to be
relevant for the generation of
sequences. They do start to kind of swim
upstream
into cognitive
states which do get into mind like
that's where this where those um
examples in the table basically go from
they're grounded in self-report or
secade um or decision or uh game theory
games or trust
games and then um latent states are
proposed which make their way kind of up
the low road towards the psychological
and the cognitive.
Mhm.
Um but maybe just to emphasize like what

I think is scientific about this chapter
u compared to what we've seen before
like if if somebody comes up with a
computer computer model using active
inference principles to explain like how
a hand reaches out and manages to
gracefully pick up a chest piece and
move it, right? like so that there's
this uh model of how the muscles
gracefully uh relax or or tense up and
they do all that. That model is uh
interesting. It could be very slick. Uh
it could be it would be it's always
impressive but it's not convincing that
that's scientifically you know really
relevant in a sense because there could
be many kind of slick ways to model
that. There could be many biological
processes that explain that but like
what this chapter is
uh portraying I think or suggesting it's
it's saying look if you have let's say a
group of individuals and you're able to
model their behavior but then you're
able to show look some of these
individuals are applying a different
strategy they're having different
beliefs they're getting different
results you know because see so that's a
very discreet great difference. You're
able to say like in these cases it's one
way, in these cases it's another way. So
scientifically like that becomes um that
for somebody like me that's very
convincing that
oh that may well be a you know the right
explanation. Maybe I'm gullible to
things like that but but um I find that

more uh yeah like let's just say we do
the rat and the teammates like the
chapter seven experiments. Mhm. So if we
just wanted to say we run 100 mice for
100 replicates each. So if we just
wanted to say okay we're making a
leaderboard which one got the highest
percentages. Mhm. We we just have that
raw data tabulatable from description or
if we wanted to say um just a simple
correlation between the temperature in
the room and the success rate which mice
did better when it was hot. These are
still basically you take two
observables, you know, we have the
observable temperature, observable
successes, and we just basically there's
our answer. So that sort of doesn't appe
appeal to any latent variables. But then
like you're saying, if we were going to
want to go beyond, so then we say, well,
how how likely do we feel that the
leaderboard would be the same on the
next Mhm. replicates or if the condition
conditions changed or we want to study
transfer learning into a different
setting.
That's where I mean why go through all
this work of making a specifying a
generative model with cognitive states
doing model inversion to get error bars
on unobservables
when if we just have the observables
let's just tabulate them. Mhm. And so
again, that goes to sort of the depth
and type of scientific account that one
seeks. If we just want
to summarize

observables, this whole apparatus could be overkill. Like if we were just trying to say we just want to know which cars are breaking the speed limit. It's just like as soon, okay, it's going over the speed limit. Boom. There's the rule-based observable. Mhm. But if you wanted to say we want to be inferring something about the attention level or the cognitive state of drivers. Mhm. And maybe want to have one model that's just speed. One is just speed and exposition or something like and they have all whole portfolios of models, portfolios of cognitive
states, noisy or patchy data. So it's a much more flexible modeling situation. Mhm. to be able to go from a custom sensor array to a custom
cognitive
unobservable
suites and then treat those unobserved
inferred
variables
as downstream as observations for a downstream process. like this could be a
risk
factor, but the but a risk
factor is never going to be a measurable feature of the world. there could be an observation that's a good proxy for risk
or something like
that, but any kind of summary
statistic or
um risk
estimator, there's just all kinds of
stuff that seems pretty obvious to want, like which one of these buildings is

going to be most likely to have this
happen,
but that will never be revealed from
just munching the data from observables
down. But even even like with a risk
factor like risk factor is generally not
a satisfactory causal explanation.
uh well I mean it's maybe helpful and
practical in some ways but but if you
really want to know the cause right like
um there's usually something direct and
I think what this is offering uh is to
say that like if different individuals
have or organisms have different um
generative
models they're going to have different
causes to their behavior
yeah again to like what makes a
scientific account what if we Okay.
Well, this person uh this building was a
high risk of falling. Mhm. Why did it
fall? It had a high risk of falling.
It's like, well, that's not really an
explanation. That might be very useful
that you were able to have a a a
calibrated risk assessments or which one
of these cars is likeliest to speed. So
that gets into the testable useful
hypothesis, but it's kind of like that's
an that's a account in the
investigator's
mind, a a utility oriented account, but
that's not even being proposed to be an
actual natural account of like why the
building fell, right? Which might relate
to like the total gravitational load and
like the supportive ability or something
like that. But the design, the design

and the materials, right?
Also, Andreas, I think you point to why
uh active inference crops up in so much
of the embodied and active mind
literature.
Uh whether the internal models are
updated
or in action is the is what's being
updated.
They can't be separated. They they
interact with each other.
Yeah. Like the the moot or the trivial
system is just
behavioral. So then there's no need to
abduce a latent state. So it's kind of
like it can't update beliefs. It can
only update its action. Yeah. But once
you start getting into anything
um another important piece is is the uh
introduction of the basian approach.
So if we go out we we do the mice
studies and like 75% of them succeed and
you say well what's your best estimate
on mice succeeding not how many mice
succeeded in your experiment but what is
your best estimate on mice succeeding in
this if you were to run it next week or
if you were to have another student run
it or something like that. The the
frequentist answer is 75. You know we
flipped the coin 75 100 times we got 75
heads. So our best estimate is 75 and
that's like the maximum likelihood
estimator. Whereas with basian
statistics we might have like a
metaanalysis of similar studies. Mhm.
And say like we have a pretty strong
belief that it's about 50%. So then on a

given day when we get a run and it gets
75 with
attention full attention to the
experimental results we could say we we
now believe that it is 75. we believe
the situation is rapidly changing and
that now mice solve it 75% of the time
or and say this is just this many sigma
it's a three sigma
events one in 700 times we expect to see
that we have now seen it one in 700
times and we're not updating from 0.5 at
all
or it could be anywhere in between with
like yep now our best posterior estimate
is
52%.
So that's the part of this whole work of
specifying the baze graph and doing this
this birectionality of like generating
synthetic data or receiving empirical
data is so that we can explicitly
include a prior belief
term because the just the munch the
description you know munch the data
style describe the
observations you don't really get a
chance to include your prior
Whereas basian statistics is basically
just including your prior as an
observable. Mhm. For the
analyst and then then making choices
about how much to weight the
prior versus the data coming in. Mhm.
Which again is that kind of reflexive
element because it's like oh yeah how
much how precision waiting on how much
to do prior updating given the

observation is the basian statistics
question. Mhm.
And descriptive statistics lends itself,
you know, not not not to weld things
together too tightly, but that lends
itself to the p value accept or reject
falsification.
Mhm. Whereas the basian epistemology and
and there's many like writings and
explorations on this lends itself to
more this continuous probability space
updating style. It's like well is the
sun going to rise tomorrow or not? you
know, whatever. Will we come to see the
sun tomorrow? Mhm. And the frequentist
could just say yes. Or, you know, I
failed to reject that it didn't. It's
like all those kinds of
expressions as opposed to just having a
very high posterior
belief, but a basian and and you could
make a basian
model that does give a discrete
answer, but if you don't constrain it
that way, it's going to stay in the
continuous probability spaces which have
more agility.
Mhm. because it's it's like okay well we
shut we shut the door on that
hypothesis. It was it was um it was
falsified. But then part of the
um sometimes facious nature of the
frequentist test is you're presented
with two
hypotheses and then either there's a
rejection or a failure to reject
one and then the failure to reject is
not the same as the

acceptance. And the rejection of one
isn't the same as the acceptance of the
other because it can all be down this
super contrived sequence
of assumptions.
So it could be like we tested to see
whether eating eggs made a difference in
cholesterol at this p value. We didn't
reject the null that it
didn't. But that's like I don't know if
it's like the contraositive or something
like that. But it's not the same as
saying that it
doesn't. It's just saying that that
study wasn't powered or or sensitized to
detect the
rejection, which is which has a huge
overlapping territory with it actually
not making a difference. Mhm. So all all
of this is kind of like in practice,
they might all give the same results. If
you're just testing for the difference
in temperature between two rooms and the
difference is big enough and clean
enough, probably a huge number of
methods are going to get you the right
um actionable information.
But this is getting towards a more
sophisticated accounting that like helps
us embody all of these nuances that that
that do come into
play. This brings us back
to the great power meaning uh
necessitating careful model
construction.
uh sort of like in our other
session, it'd be worthwhile. I'm looking
for a course on modeling. Uh the

philosophy of models.
Well, I would talk about the model I'm thinking of that's uh uh this chapter kind of provoked me to think about uh because I'm I like to model human experience. Um I've been working on that for decades and and so our active inference is um attractive um and um but and so active inference is quite universal but but maybe not completely. So an example of the type of problem I thought I would do I would like to know what are the languages by which uh for a human being uh things come to matter or u let's say a language for how does meaning arise or how does a person believe like an event happened you know I think that there's maybe some basic cognitive languages for that so I needed data for that so the data I thought I would study uh is u triangle geometry uh that you can have triangles of different shapes and then you have these concepts of triangle centers which could be the simp a simplest one is what's called an in center. You draw a circle inside a triangle and you make the circle as big as possible. The center of that circle will be called the in center of the triangle. But you could do it with a center on the outside. And there's all these ways that you could um make points given a triangle like you know you look at the medians you you intersect them things. So but there's not that many ways. And I suspect if you if I study that and there's an encyclopedia of triangle centers and if I study how it

builds it builds a language a language
like this or maybe any language will
probably
have six underlying actions uh which
relates to let's say six cases uh Daniel
gave a um a talk about u cases in
language let's say so something like
that these type of prepositional ways of
thinking um that nouns are let's say
used in a language like Lithuania or
Russian or so on or Latin. So um the
point being that okay as far as active
inference goes it's kind of interesting
that it's possible to think of there is
this kind of like coordination between
two types of action. One is let's say a
physical continuous action like making a
circle bigger. The other one is saying
there's like a conceptual point to say I
want to have a center of a circle. So
you do actions like you make a circle
bigger. It's not positioned well. You
kind of bump it over. You keep doing you
get you do a chain of actions. You end
up with something like that's a center.
Let's say that could make sense for a
generative model saying this circle
needs a center that I could focus on.
And maybe you can redo it for different
triangle shapes and do different types
of triangle. You get this whole kind of
activity for language. The problem with
that based on this chapter is that yeah,
but is that ever going to be science?
you see like is that is that ever going
to solve a scientific issue because
you're just um studying um you know
these triangles it's not in a certain

sense it's not really data from the real
world in terms of experimental data on
the other hand it kind of is because you
have this whole you know encyclopedia of
triangle centers people came up with
what does active inference say about an
encyclopedia like that like how to make
sense of it like so if there was a
modeling language or approach that could
go into that data and say hey this is
how to un this is what to do to
understand it as a language now so the
one thing I wanted to just conclude with
is that um what this chapter got me
thinking though is that in order to make
it to the next level where it's kind of
more observable or it's more
meaningful you kind of need the system
or the game to um maybe like be
communicating with another individual.
So like let's say there's two people
playing this game of u the triangle or
maybe like you know you're just
communicating with yourself you're
replaying you know you're trying to get
good at this you're trying to make sense
of you know what is a triangle uh what
helps me understand triangles see so
when you model that and how those games
are related and I think the reason being
that like in the science of this what's
impressive is when you have two
different people with two different
generative models and they get different
behavior so if you do that in a single
person like with this type of game
saying well you can go into different
modes or have different beliefs or they

can change or you can do that in
communicating with somebody else then I
think that becomes the kind of thing
we're trying to be able to model
yeah very interesting yeah Stephen
please I think that's the essence of the
human mind that there are two generative
models that are compared with each other
there's an online sensory input and then
there's the
internal representation of that sensory
input and those get compared with each
other and it's that comparison I
think in in the higher level cortical
areas
that is this really
complex motor behavior that we call
thought sense of self and motivation. I
I I wonder so there's that. Uh I I
wanted to ask if you
were
using meaning in the sense of concept
like the formation of meaning is the
formation of concepts. I think there's a
whole literature in what concepts. Well,
of course, there's literature in what
meaning is also, but I'm always
approaching this from the standpoint of
the actual architecture of the brain and
the body and how this is
implemented uh you know neurons or
basian units themselves and then they
form basian circuits and and it's the
dynamic processes you know with these
frequenc of uh uh these activations with
different frequencies that allow signals
to get through or not, you know, that
amplify or cancel

signals. And you know, it's it's so hard for me to constantly translate basian terminology like beliefs into a structural terminology like neural circuit.

So um what I find very helpful what you said was like that you see generative models at both levels and I really hadn't thought about that but maybe that's the way to go. Um I have this model of the three minds I call it. Uh so there there's a mind that knows the answers uh which would be what you call like the sensory um and it knows them unconsciously but then you have a mind that uh uh does not know but it consciously does not know. So it's just like full of questions. It's full of variables like every word is a question like so like the word horse really means like what is a horse and then the other mind gives you the answer what the horse is maybe gives you an image or something. So so in terms of meaning arising it's like basically you have these two minds on autopilot you know like one is just um one is just feeding you all kinds of imagery like you're watching television the other one's just kind of chattering away you know and they just do their own thing. Um, but you have a third mind that's investigatory that tries to connect them two, you know, with each other to kind of like um balance them so that the mind that knows and the mind that does not know, they're able to say the same things in their different ways. Cuz like

the mind that knows, it's like a 100 billion neurons are telling you one thing, but there's like a 100,000 concepts that that same information is being reduced. And now so it's like this distinction between the brain and the mind, you know, like that is that neural well that's maybe not a very helpful way to look at it. It's just some kind of conceptual structure out there. Um but there they need to be linked. So there's a third mind that kind of helps to point punctuate or um you know adjust ratios, parameters, whatever to kind of uh emphasize like to weave them together. So on the one point of view like with the conceptual mind there's a certain point where like it's babbling and all of a sudden yeah but this is meaningful and that is meaningful for this reason and for that reason like for example this point at this circle is going to be meaningful that we're building. Uh whereas like the the points on the circle aren't meaningful at all but in terms of mattering like it's it matters to get the circle to the maximal size that matters. That's a task to perform. So, I like that little language because you can see the difference between the two. Yeah. I think the third mind falls out of the other two. It's a consequence of the interaction between I really like your framing there. Yeah. And so, it's both a consequence, but then maybe it's steering it. You see? So, like we have a will. It's deliberate. It's cognizant.
So, it's a it's ambigu it's ambiguous,

right? Like which is it in the process?
It
changes. One could say it changes the
synaptic weights but the uh uh there
there's nothing static. Everything is
constantly changing and when they meet
the interaction changes itself in the
process of interacting. Think of I
always think of it in terms of a
syninnapse changes itself in the very
action of transferring neurotransmitter.
the vesicles have proteins on them that
merge with the synaptic membrane and now
it's different than it was before. Yeah.
So is that is that a helpful way to look
at neuro is is active inference and
generative model I mean generative
models are they a helpful way of looking
at the neuronic neural systems
themselves.
It's a it's rich enough to apply to the
neural systems themselves. Absolutely.
Yeah. It's very it's quite appropriate I
think. uh there I think there that's
where it's really helpful to look at the
SPM work which is a whole toolkit
ranging from
like spike and fire models of single
neurons to neural mass probability
models to dynamic causal modeling all
within this like ultra chapter 9
empiricism with the sex fusion of the
EEG fMRI data
Um so this and then in a way the the
scientists
model can be seen as like a symbolic
second mind of the primary embodiment of
the mouse. But let's say it was like an

alien creature and it was it was or it's
an unknown creature but it h it has an
embodiment. It has a firstness. Mhm. And
then this is a symbolic
projection onto
that firstness from a pretty small
diagrammatically
defined graphical toolkit. Mhm. And then
there's a to your point about the
communication then there's a two agents.
It's kind of like that. Then the minds
is kind of like the the experimental
setup and the experimentter.
Mhm. And then there's the investigation
is in the
interplay. And then you talked about
communication of generative models
across agents. That's like the
synchronic like joint. That's that's
like the space between two at or or more
at once. And then there's um replay and
and pre-play
which is like
diarronic but that so that's like one
self but but it's like is it a different
self right? If it's a but you know just
legally speaking it's the same person.
Mh. But then that's like also kind of
like a space
between some extent
actuality which is like the 100 million
neurons. Mhm. And then some
symbolic
condensation and then some something
happens in the space between which in
the basian updating
moment is given a causal power.
Or you could just take the the basian

posterior and just it could be treated
as
epifenomenal like you could just
continue to to collide streams of priors
and observations.
Mhm. And then just slough off the
posteriors.
What do you mean by a phenomena? What
does that mean? Like you'd calculate a
posterior and discard it. Like the the
posterior is never given a causal
re-entry into the system. Oh, okay. Like
what what people arguing that that
consciousness is
epifenomenal, which is to say like it
arises but there's no causal arrow back
like it's like an it's like a side
effect but not really relevant. It's not
a participant. Yeah. And then the
question is like you know without you
know dart and all that but how do you
have something that re-enters causally
is the third mind just like the sparks
flying from the interplay but then but
do those sparks you know c can they or
you know you know if weather whether
it's that all those things that that you
know do they re-enter to modify either
neural activity or
symbolic activity
and and so I'm writing this paper to
about this model and and u some of the
things I write about uh and I quote the
textbook uh but for example there's an
interesting paper about a window that
says you know you have pre you have a
stimulation of the brain let's say you
have preconcious uh uh time but at let's

say from 250 milliseconds to 700 milliseconds there's this rather long window of conscious axis is what they call and what then what they noted with the paper they did different situations where you could do task related or uh taskfree uh you know you're listening to music let's say maybe you're supposed to be hearing look hearing for something or maybe you're not but but they they look at the conscious axis and what they notice is that um the window stays open for longer than you would think like there's room for playing around of the signal there's room for it to settle down so with regard to the three minds then that's very compatible in the sense that your first mind these this preconscious stage can suggest candidate action uh sensations it can give you it can give you information like that then your conscious mind is globally trying to figure out you know how to integrate this what to do but the third mind it's like acting as a break it's saying wait you know don't rush don't rush I'll tell you when to hardwire it when it's all right so the point being that the prolongation suggests that there could be something that's prolonging this, you know, like that's that's that's ex, you know, fighting against early u early resolve resolution. But furthermore, you can have an idea go into your brain before you're conscious of it of what you're going to do. It's just a candidate, you know, decision. So you get this paradoxical thing that oh what

you thought you were going to do you already knew quote unquote before you but actually it was just a candidate you know that got confirmed. I think that's the way to read it. I like that which then is affected by experience. the more that happens, the more likely it's Yeah. But all three zones like the meaning changes as you go through those three zones of you know it was a candidate it be got kind of like worked out and then it was affirmed let's say and and I think that points to a threshold effect uh where if there is this practiced development of openness to intention then if consciousness can be ever be considered as anything other than epiphenomena it is in the realm of intention.
uh creating as you say uh a bias toward a particular candidate class of yeah you can have a bias right and so that's a way of self affecting yourself is to have
biases and so so I quote chapter five uh about the basil ganglia because I say there's a lot of evidence I think mielchrist Ian Mcielchrist wrote a book about left hemisphere versus right hemisphere so you get a lot of this you know a champion and for both but I'm saying it's deeper than that. If you go to chapter 5 about the basil ganglia uh even in arthropods uh you have this uh uh distinction between u releasing and allowing the action as opposed to not allowing it. There's this you know you have to play with both and I think that

that's a sign of consciousness that uh
you know you're able to do that. Uh and
so dopamine is what triggers that and
dopamine goes way back. So that's just
the mechanism. Another one I cite is uh
uh immune systems. You have two
versions. You have um the um the one
that's like inborn, innate I guess they
call it, then the one that is able to
learn. And so the question is how do
they not
conflict? So I think there's got to be
some kind of harmony between them,
harmonizer between those two immune
systems. You talk about another
principle of biological function.
uh the the best control system is one
that is both excitatory and inhibitory
in constant
right you want both right and so I I
have this quote I I I said this thing
the second mind it avoids the territory
because there's things you can't learn
from you know you'll just die so you
want to avoid the territory and embrace
the map uh and so that's why you want to
for things you know you you know like
the unlucky mind there's a mind when
things are unlucky you don't want to try
your luck we have a lucky mind and an
unlucky mind so okay thank you we'll end
it
there see you